UNIT 10 · STANDARD 6.G.2

Volume of Rectangular Prisms

Lesson 10-2

Name: ______________ Date: ______________ Class: ______________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

A box $2 \times 1.5 \times 1 = 3$ ft³ — it holds 3 cubic feet of stuff

Volume

How much space is inside a solid shape.

A tissue box, a fish tank, a brick — all rectangular prisms

Rectangular prism

A solid box shape with six flat rectangle sides.

1 ft³ = a cube that is 1 foot on every edge

Cubic units

The units used to measure space inside, like cubic inches.

A box with dimensions $4 \times 3 \times 2$ means $l = 4$, $w = 3$, $h = 2$

Dimensions

How long, how wide, and how tall a shape is.

Cut a cereal box along its edges, unfold it flat — that is the net

Net

A flat shape that folds up into a solid.

If the base is $6 \times 4 = 24$ cm² and $h = 3$, then $V = 24 \times 3 = 72$ cm³

Base area

The area of the bottom of a solid. Volume = base area × height.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

1 Notice

Your class is building a time capsule to bury on school grounds.

2 Key idea

I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area × height.

3 Words I'll use

Volume

Rectangular prism

Cubic units

Dimensions **Net**

Base area

🤔 Think About It

- What are the three measurements of the time capsule?
- What does volume measure — length, area, or space inside?
- How is volume different from area?

👀 Watch

Watch your teacher model one example. Jot what you see.

🤝 We try

Solve the next one together as a class.

✏️ You try

Now try one on your own.

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Your time capsule is 2 ft long, 1.5 ft wide, and 1 ft tall — one edge is a fraction. Does having a fractional edge change how you find the volume? Why or why not?

Level 1 support

Start here: You still multiply length × width × height even when an edge is a fraction or decimal.

💡 **Need a hint?**

Sentence starters (optional)

A fractional edge ____ change the method because I still multiply ____ × ____ × ____.

Una arista fraccionaria ____ cambia el método porque sigo multiplicando ____ × ____ × ____.

The volume of the capsule is ____ ft^3.

El volumen de la cápsula es ____ ft^3.

Word bank: volume | dimensions | rectangular prism | base area | cubic units

Listen for: *Students explain the formula stays the same ($V = l \times w \times h = 2 \times 1.5 \times 1 = 3\ ft^3$) and that a fractional edge just means multiplying with a fraction or decimal.*

Level 2

How could you use base area × height to find the same volume, and why does that give the same answer as l × w × h?

- Base area × height gives the same volume because base area is ____.
- First I find the base area ____ × ____ = ____, then multiply by the height ____.

EXPLORE

Capsule A ($2 \times 1.5 \times 1$) and Capsule B ($3 \times 1 \times 1$) both have a volume of 3 ft^3. How did your multiplication produce the same volume from different dimensions?

Level 1 support

Start here: Different dimensions can multiply to the same product, so the volumes match.

💡 **Need a hint?**

Sentence starters (optional)

Both capsules have V = ____ ft^3 because ____ × ____ × ____ = ____ × ____ × ____.

Ambas cápsulas tienen V = ____ ft^3 porque ____ × ____ × ____ = ____ × ____ × ____.

The dimensions are different but the ____ is the same.

Las dimensiones son diferentes pero el ____ es el mismo.

Word bank: volume | dimensions | rectangular prism | base area | cubic units

Listen for: *Students show $2 \times 1.5 \times 1 = 3$ and $3 \times 1 \times 1 = 3$, and explain that the same product means equal volume even with different shapes.*

Level 2

If two capsules have the same volume but different dimensions, which would you choose to store long, flat objects, and why?

- I would choose Capsule ____ because its dimensions ____.
- Same volume does not mean same usefulness because ____.

CONNECT

A fish tank is 24 in long, 12 in wide, and 16 in tall, and one gallon takes about 231 cubic inches. Talk through how you would find how many gallons it holds.

Level 1 support

Start here: Find the volume with V = l × w × h, then divide by 231 to convert to gallons.

💡 **Need a hint?**

Sentence starters (optional)

The tank's volume is ____ in³ because 24 × 12 × 16 = ____.

El volumen del tanque es ____ in³ porque 24 × 12 × 16 = ____.

It holds about ____ gallons because ____ ÷ 231 ≈ ____.

Contiene unos ____ galones porque ____ ÷ 231 ≈ ____.

Word bank: volume | cubic units | rectangular prism | dimensions | base area

Listen for: *Students compute 4608 in³, then divide by 231 to get about 20 gallons, and explain that dividing converts cubic inches into gallons.*

Level 2

Why does the answer come out to about 20 gallons instead of an exact whole number, and how would you decide whether to round up or down for filling the tank?

- The answer is not exact because ____.
- To fill the tank safely I would round ____ because ____.

REFLECT

A classmate says a $5 \times 3 \times 2.5$ m prism has volume $5 \times 5.5 = 27.5$ m³. How do you find and explain their mistake?

Level 1 support

Start here: They added $3 + 2.5$ instead of multiplying; the correct volume is $5 \times 7.5 = 37.5$ m³.

💡 **Need a hint?**

Sentence starters (optional)

The mistake is that they ____ instead of multiplying 3×2.5.

El error es que ____ en lugar de multiplicar 3×2.5.

The correct volume is ____ m³ because $3 \times 2.5 =$ ____, then $5 \times$ ____ $=$ ____.

El volumen correcto es ____ m³ porque $3 \times 2.5 =$ ____, luego $5 \times$ ____ $=$ ____.

Word bank: volume | dimensions | rectangular prism | base area | cubic units

Listen for: *Students identify the added-instead-of-multiplied error, compute $3 \times 2.5 = 7.5$, and conclude the correct volume is 37.5 m³.*

Level 2

What general rule about the volume formula does this mistake break, and how could a quick estimate have caught it?

- The mistake breaks the rule that volume always uses ____.
- An estimate would catch it because ____.

Guided Examples

Example 1

What is the volume of a rectangular prism with l = 4 in, w = 3 in, h = 5 in?

Solution: $V = l \times w \times h = 4 \times 3 \times 5 = 60$ cubic inches.

Answer: A. 60 in^3

Example 2

A box has a volume of 36 cm^3. Its length is 6 cm and width is 3 cm. What is the height?

Solution: $V = l \times w \times h \rightarrow 36 = 6 \times 3 \times h \rightarrow 36 = 18h \rightarrow h = 2$ cm.

Answer: A. 2 cm

Example 3

How is volume different from area?

Solution: Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).

Answer: A. Volume measures 3D space (cubic units); area measures 2D surface (square units)

Write About the Math

The Writing Revolution

I can explain my work using the words volume, rectangular prism, dimensions, and base area.

1. Kernel Sentence

subject + verb

Model: Volume is how much space is inside a solid shape.
Volumen es cuánto espacio hay dentro de una figura sólida.

Write a kernel sentence about volume. Use a subject and a verb.
Escribe una oración base sobre volumen. Usa un sujeto y un verbo.

___

___

2. Sentence Expansion

because · but · so

Kernel: Volume matters in math
Volumen importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because *porque*	**Volume matters in math because ____.** *Volumen importa en matemáticas porque ____.*
but *pero*	**Volume matters in math, but ____.** *Volumen importa en matemáticas, pero ____.*
so *entonces*	**Volume matters in math, so ____.** *Volumen importa en matemáticas, entonces ____.*

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about volume.
Di un hecho verdadero sobre volume.

Volume ____.

Question
Pregunta

Ask a question about volume.
Haz una pregunta sobre volume.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about volume.
Muestra entusiasmo sobre volume.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with volume.
Dile a un compañero qué hacer con volume.

First, ____ .
Primero, ____ .

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4. Explain Your Reasoning

use a sentence starter

A fractional edge ____ change the method because I still multiply ____ × ____ × ____.
Una arista fraccionaria ____ cambia el método porque sigo multiplicando ____ × ____ × ____.

The volume of the capsule is ____ ft³.
El volumen de la cápsula es ____ ft³.

Both capsules have V = ____ ft³ because ____ × ____ × ____ = ____ × ____ × ____.
Ambas cápsulas tienen V = ____ ft³ porque ____ × ____ × ____ = ____ × ____ × ____.

Try It

Solve on your own. Check the answer key when you are done.

1. How is volume different from area?

A. Volume measures 3D space (cubic units); area measures 2D surface (square units)

B. Volume uses square units; area uses cubic units

C. Volume only applies to cubes; area applies to all shapes

D. There is no difference

Show your work:

2. Two boxes: Box A is 10 × 5 × 4 inches. Box B is 8 × 6 × 5 inches. Which has more volume and by how much?

A. Box B by 40 in^3

B. Box A by 40 in^3

C. Box B by 60 in^3

D. They're equal

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A toy company ships action figures in boxes that are $4 \times 3 \times 6$ inches. A shipping crate is $24 \times 12 \times 18$ inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

Sentence starter: Each toy box has a volume of ____ in^3. The crate has a volume of ____ in^3. I can fit ____ boxes because ____. I checked by ____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 7$ ft, $w = 2$ ft, $h = 3$ ft. What is the volume?

A. 42 ft^3

B. 24 ft^3

C. 42 ft^2

D. 12 ft^3

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. Volume measures 3D space (cubic units); area measures 2D surface (square units) — *Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).*
2. **Try It 2:** A. Box B by 40 in^3 — *Box A: $10 \times 5 \times 4 = 200$ in^3. Box B: $8 \times 6 \times 5 = 240$ in^3. Box B is bigger by 40 in^3.*
3. **Exit Ticket:** A. 42 ft^3 — *$V = 7 \times 2 \times 3 = 42$ cubic feet. Remember: volume uses cubic units (ft^3).*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Volume is how much space is inside a solid shape.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).